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ABSTRACT

Additionally, data preparation is the most resource-intensive stage in the Data Science lifecycle and can account for 80% to the entire project duration. However, the use of LLMs has been shown to turn the tide, as it can significantly aid data cleaning. However, the model's sensitivity to question structure is of great interest in the broader field of Prompt Engineering.

Moreover, the primary issue this work addresses is the vulnerability of reasoning-based prompt designs to real-world data with substantial noise. Generally, traditional prompt design methods are less effective when faced with typographical errors or formatting inconsistencies. This is highly relevant to real-world contexts that often involve less-than-clean data. It focuses on whether such 'reasoning' prompt designs will degrade Entity Resolution.

To tackle the problem, the proposed dissertation adopts and implements the Evolutionary Prompting System (EPS) to incorporate Genetic Algorithms (GA) to optimize the prompts automatically. The EPS treats the prompts as 'chromosomes' to maximize cleaning accuracy with the fewest tokens.

- Building a “Hard Mode” Benchmark: The original Fodor 's-Zagats dataset was augmented to produce 560 different profiles to model extreme OCR errors and case swap inconsistencies.

- Algorithmic Evolution: Utilizing GPT-4O as a mutation operator to repeatedly rewrite and optimize prompts based on their fitness levels (fuzzy matching accuracy).

- Benchmarking: Comparing our Evolutionary System with industry-standard Zero-Shot and Chain-of-Thought benchmarks.

Contribution and Key Outcomes: The results provided a statistically significant and counter-intuitive contribution to the field of Automated Data Cleaning. The experiment revealed a catastrophic failure mode in the standard **Chain-of-Thought approach, which achieved 0.00% accuracy**. Qualitative analysis showed that "reasoning" on garbage data caused the model to hallucinate complex justifications and violate output schemas.

In contrast, the **Evolutionary System achieved 70.00% accuracy**, significantly outperforming the Zero-Shot baseline (64.00%) and recovering entirely from the CoT failure. The study concludes that, for noise-intensive data preparation tasks, evolutionary robustness yields superior results than **reasoning-based complexity**. This research offers a viable, low-resource methodology for automating the data quality stage of the CRISP-DM lifecycle without relying on brittle, hand-engineered prompts.

ACKNOWLEDGEMENTS

I am grateful to my parents and lecturers for their support in my studies and in my life.

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CHAPTER 1: INTRODUCTION

The rapid growth of unstructured data in today's business environment has shifted data quality management from a maintenance process to a necessity. Although Large Language Models (LLMs) show immense potential in Natural Language Understanding, their utility for automated data cleaning, particularly in Entity Resolution, remains limited due to their sensitivity to question formulation and input noise. The proposed dissertation aims to address the above drawbacks by formulating an Evolutionary Prompting System (EPS).

1.1 Background

Conventionally, the problem of Entity Resolution (ER) — the ability to identify and merge duplicate data that refer to the same actual entity — has been addressed by brittle rule-based systems or manual annotation. The recent advent of Large Language Models (LLMs) has precipitated a paradigm shift in this area. The GPT family of LLMs can analyze dirty data without necessarily requiring a trial-by-fire large-scale dataset. "Prompt Engineering"—the art of constructing language prompts capable of directing LLMs — is now an established subfield within Computer Science, offering a potential solution to the automatic handling of the data cleansing cycle.

State of the Art and Research Gap

The current "state of the art" in Prompt Engineering strongly emphasizes reasoning-heavy approaches. In fact, methods of "Chain-of-Thought" (CoT) prompting, which involve having the model explain its reasoning process, are currently the norm for improving performance on complex logic and arithmetic problems. Other methods, including "Zero-Shot" and "Few-Shot" prompting, have achieved some success in general NLP tasks.

However, there lies a crucial gap in the use of such methods on real-world data quality tasks. Although complex reasoning chains (CoT) perform exceptionally well in the well-structured academic testing environment, there are indications that they may be vulnerable when applied to the "unstructured and noisy" data of a business environment. The use of prompt complexity in the presence of data noise has been neglected in the existing literature. Moreover, the use of automatic prompt engineering (APE) in robust entity resolution remains underexplored.

However, a critical gap remains in applying these techniques to real-world data quality problems. While complex reasoning chains (CoT) excel in structured academic tests, anecdotal evidence suggests they may introduce fragility when applied to unstructured, noisy business data. Existing literature largely overlooks the interaction between *prompt complexity* and *data noise*. Furthermore, while **Automatic Prompt Engineering (APE)** and evolutionary algorithms have been proposed to optimize prompts for mathematical reasoning, their application to the domain of robust Entity Resolution remains under-researched.

Motivation & Significance: The motivation for the current study stems from the observation that "smarter" prompts do not always yield "better" prompts when the prompting inputs are degraded. If complex reasoning models, including the use of CoT, fail due to the effects of OCR errors or typos, the current industry standard for prompts warrants reconsideration.

This dissertation addresses the need for a robust, automated approach to optimize the process for handling dirty data prompts. This connection between Evolutionary Computation and Prompt Engineering will inform the development of a computerized approach to evolve noise-resistant prompts. Among the advantages of this proposed approach is its significant impact on the Data Science field, particularly by reducing the manual process of optimizing prompts.

1.2 Research aim and objectives

The goal of this project is to develop an Evolutionary Prompting System (EPS) that uses genetic algorithms to automatically optimize Large Language Model prompts, providing data scientists with a reliable framework for Entity Resolution on unstructured, noisy datasets.

Objectives;

- To review the literature regarding Entity Resolution, Chain-of-Thought prompting, and the application of Evolutionary Algorithms in NLP.
- To construct a "Hard Mode" benchmark dataset by synthetically augmenting Fodor 's-Zagats standard with OCR noise and typographical errors.
- To develop the Evolutionary Prompting System in Python, using Large Language Models as mutation operators to refine prompt effectiveness.
- To experimentally benchmark the system against industry-standard Zero-Shot and Chain-of-Thought baselines to quantify performance gains.
- To critically evaluate the limitations of reasoning-based prompting on dirty data and discuss the implications for automated data cleaning pipelines.

1.3 Research approach

This dissertation employs a quantitative experimental design, mainly organized around the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, to achieve the aforementioned goals. Empirical benchmarking is the primary focus of the iterative research design. The approach is divided into three primary phases:

- Data Preparation and Augmentation: The public Fodor's Zagat restaurant dataset is used in this investigation. A novel noise injection process was developed to artificially introduce optical character recognition (OCR) flaws, typographical inconsistencies, and formatting variations into the dataset to replicate the challenges of real-world "dirty" data. As a result, a "Hard Mode" benchmark was developed to rigorously stress-test the models.
- Algorithm Development: Python and the OpenAI API were used to create an Evolutionary Prompting System (EPS). This system treats natural-language prompts as evolving genotypes that optimize themselves across successive generations to enhance data-cleaning accuracy by applying concepts from evolutionary computation (selection, crossover, and mutation).
- Comparative Analysis: Two industry-standard baselines, Zero-Shot Prompting and Chain-of-Thought (CoT) reasoning, are used to assess the evolutionary system's effectiveness. Fuzzy matching metrics (RapidFuzz) are used to quantify performance by comparing the output's semantic accuracy to a ground-truth key.

Ethical Considerations

It is crucial to note that this study is fully computational and uses anonymized, publicly accessible datasets. The procedures for data collection, experimentation, and result evaluation did not involve human subjects.

Therefore, special precautions for the safety of human subjects were not necessary for this investigation, even if conventional ethical norms involving data integrity and plagiarism were properly adhered to.

1.4 Dissertation outline

The remaining portion of this dissertation is divided into six chapters that logically move from the theoretical underpinnings to the system's empirical assessment and critical debate.

Chapter 2: The literature review provides a comprehensive analysis of the current status of data quality and automated cleaning research. It critically analyzes the evolution of Entity Resolution (ER) procedures, from traditional schema matching techniques to modern Large Language Model strategies. This chapter explicitly examines the "state of the art" in prompt engineering, highlighting the limitations of reasoning-based methods (such as Chain-of-Thought) when applied to noisy, unstructured data. It also explores the application of evolutionary algorithms in data mining to provide the theoretical underpinning for the proposed method.

Chapter 3: The Research Approach (Methodology) describes the experimental design and technical implementation of the project. The synthetic data augmentation pipeline that simulates typographical irregularities and real-world OCR issues is described, as is the process for creating the "Hard Mode" benchmark dataset. The design of the Evolutionary Prompting System (EPS) and the genetic algorithm operators (mutation, crossover, and selection) employed to optimize natural language prompts are also described.

Chapter 4 presents the study's empirical results through data analysis. It offers a numerical assessment of the system's effectiveness on the 560-record supplemented dataset. This chapter presents a statistical comparison between the proposed EPS and industry-standard baselines (Zero-Shot and Chain-of-Thought). It illustrates the evolutionary development of the prompt populations over eight generations.

Chapter 5: Chapter 4's results are critically interpreted in the discussion. In particular, it addresses the reported failure of the Chain-of-Thought baseline by examining the trade-off between quick complexity and robustness. The chapter provides a qualitative investigation of specific error scenarios, discusses why "reasoning" tactics may degrade performance in low-quality data contexts, and situates these findings within the broader field of data science.

Chapter 6: The study's ramifications and significant scientific contributions are summed up in the conclusion. It assesses the extent to which the study's goals and objectives were achieved, acknowledges the limitations of using synthetic data, and suggests directions for further investigation, such as applying the evolutionary framework in privacy-sensitive domains, such as banking and healthcare.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

This part takes a rigorous look at all the research on how evolutionary computation, large language model (LLM) reasoning, and automated data cleaning work and connect to each other. This review shows how Entity Resolution (ER) has evolved, starting with older rule-based methods and moving to newer approaches such as Deep Learning and Generative AI. We'll check out how ER approaches were developed in Section 2. This change places the focus on Large Language Model solutions, and, of course, that comes with its own set of problems, mainly ensuring they work well and don't cost a fortune. Section 2 examines prompt engineering in detail. Let's discuss the issues with AI reasoning methods, such as Chain-of-Thought, mainly when they're used with messy, unorganized information. Finally, let us look at Section 2. Chapter 3 examines how Natural Language Processing and Evolutionary Algorithms interact. It establishes the basic theory for the Automatic Prompt Optimization (APO) system I propose in this paper.

2.1 Evolution of Entity Resolution Techniques

2.1.1 The Importance of Data Quality

One of the significant obstacles to implementing reliable AI systems is data quality. It is commonly known that "dirty data" can significantly impair downstream model performance. Misspellings, duplicates, and uneven formatting typify this type of data. The primary method for resolving this problem is Entity Resolution (ER), which is the process of identifying and linking data that refer to the same real-world entity. However, as data volume increases, the cost of traditional manual labeling has become prohibitive, underscoring the need for automated alternatives.

2.1.2 From Rules to Deep Learning

In the past, ER relied on symbolic, rule-based systems that employed string similarity metrics such as Jaccard similarity or Levenshtein distance. These systems were interpretable but were unable to capture semantic nuance and required extensive domain-specific rule engineering. A significant paradigm shift occurred with the introduction of Deep Learning. Recurrent Neural Networks (RNNs) and attention mechanisms were used by systems such as DeepMatcher (Mudgal et al., 2018) to learn fuzzy matching patterns from labelled data, much outperforming conventional techniques. Ditto (Li et al., 2020) then improved accuracy on benchmark datasets by using pre-trained language models (PLMs) like BERT to represent entity matching as a sequence-pair classification problem.

2.1.3 The Generative AI Era (LLMs)

Generative Large Language Models (LLMs) like GPT-4 and Llama-3, which have the exceptional capacity to execute ER in a "Zero-Shot" context without task-specific training, have been the focus of the field most recently. By leveraging their extensive semantic knowledge, LLMs can achieve state-of-the-art performance on data preparation tasks, such as error detection and schema matching, as reported by Peeters and Bizer (2023) and Jellyfish (Zhang et al., 2024).

2.1.4 Critical Evaluation and Research Gap

- Despite their promise, LLM-based approaches face two critical limitations:
- Cost and Latency: According to Li et al. (2024), the "pay-per-token" paradigm of commercial LLMs makes processing big datasets unaffordable, requiring effective, quick methods.

- Fragility to Noise: Although LLMs are excellent at clear intellectual tasks, noise can impair their ability to reason. According to a recent study on "Noisy Rationales" by Zhu et al. (2024), Chain-of-Thought (CoT) prompting is highly susceptible to input perturbations, with accuracy decreasing by more than 20% when false or irrelevant material is added.

This highlights a critical lacuna in the literature: although "reasoning" prompts (CoT) are frequently touted as a universal performance booster, little is known about their effectiveness in "dirty" data contexts, where the reasoning chain itself may be hallucinated. To close this gap, this dissertation proposes an evolutionary method for identifying strong cues that avoid these errors.

CHAPTER 3: RESEARCH APPROACH

The methodological framework and technical implementation for achieving the research goal of developing a reliable automated data-cleaning system are described in detail in this chapter. The study employs a quantitative experimental design adapted from the Cross-Industry Standard Process for Data Mining (CRISP-DM) lifecycle. In Section 3.1, the computational tools used are defined, and the experimental methodology is justified. The setup and data collection for the "Hard Mode" benchmark are covered in Section 3.2, which also explains the synthetic noise-injection pipeline used to replicate actual faults. The Evolutionary Prompting System (EPS) architecture is described in Section 3.3, along with the genetic algorithm operators—selection, crossover, and mutation—that are employed to enhance the Large Language Model prompts iteratively.

3.1 Methodology Selection

3.1.1 Justification of Research Approach

The methodological design of this project is guided by two major conclusions derived from the results of the Literature Review (Chapter 2). First, a rigorous quantitative approach to measuring resilience is required, as Large Language Models (LLMs) are susceptible to input perturbations. Second, Prompt Engineering's "trial-and-error" approach is more in line with iterative experimental design than with linear software development. To empirically compare the proposed Evolutionary Prompting System (EPS) with established baselines, this study employs a quantitative experimental design.

3.1.2 Selection of Data Science Framework (CRISP-DM)

Several common data science approaches, such as KDD (Knowledge Discovery in Databases), SEMMA (Sample, Explore, Modify, Model, Assess), and CRISP-DM (Cross-Industry Standard Process for Data Mining), were assessed to organize the experiment.

- SEMMA is essentially an SAS-proprietary sequence appropriate for statistical analysis rather than iterative system development; KDD places a strong emphasis on pre-processing and pattern finding, frequently lacking the deployment-centric focus necessary for system building.
- The best framework for this dissertation was determined to be CRISP-DM. The evolutionary feature of this research, in which the "Modeling" and "Evaluation" phases are repeated across several generations to refine the prompts, is well accommodated by its cyclical structure (see Figure 3.1).

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3.1.3 Application of CRISP-DM to this Project

The CRISP-DM lifecycle is applied to this project as follows:

1. Business Understanding: The objective is defined as minimizing the cost and error rate of automated data cleaning in enterprise environments.
2. Data Understanding: Analyzing the schema and quality of the Fodor 's-Zagat's benchmark dataset to identify common entity resolution challenges.
3. Data Preparation: This phase involves the critical "Hard Mode" augmentation (detailed in Section 3.2), where synthetic noise and OCR errors are injected to stress-test the models.
4. Modeling: The development of the Evolutionary Prompting System (EPS), which uses Genetic Algorithms to iteratively "evolve" natural language prompts.
5. Evaluation: Benchmarking the evolved prompts against Zero-Shot and Chain-of-Thought baselines using fuzzy matching metrics (RapidFuzz).
6. Deployment: (Theoretical) Proposing how the optimized prompts can be integrated into production ETL pipelines.

3.1.4 Data Access and Ethical Considerations

- Data Access: This research utilizes the Fodors-Zagats dataset, a publicly available benchmark widely used in the Entity Resolution literature (e.g., DeepMatcher, Magellan). The data was accessed via the open-source Magellan Data Repository and requires no special permissions or non-disclosure agreements.
- Ethical Considerations: As this project relies entirely on public, anonymized bibliographic data, it does not involve human participants or personally identifiable information (PII). Therefore, no ethical approval for human subjects was required. The primary ethical consideration was ensuring the responsible use of the OpenAI API by strictly adhering to its usage policies on automated content generation.

3.2 Data Construction: The "Hard Mode" Benchmark

A typical clean dataset was insufficient to thoroughly stress-test the robustness of Large Language Models (LLMs). To replicate the entropy of actual enterprise data, a unique "Hard Mode" benchmark was built.

3.2.1 Source Data:

The study makes use of Fodor 's-Zagats dataset, which is a collection of bibliographic restaurant records from the Magellan Data Repository.

- **Schema:** [Name, Address, City, Cluster_ID]

- **Base Volume:** 112 matched pairs.

3.2.3 Data Augmentation:

Data augmentation was employed to ensure statistical significance and prevent overfitting. Third. Using distinct random seeds, each original record was duplicated and corrupted five times (Augmentation Factor = 5).

- **Final Dataset Size:** 560 unique test cases.
- **Split:** 70% Training (used for Evolution), 30% Testing (held out for Final Exam

3.3 The Evolutionary Prompting System (EPS)

An Evolutionary Algorithm (EA) used in Prompt Engineering serves as the

central mechanism of this study. This system optimizes semantic natural language strings, in contrast to conventional GAs that optimize binary strings.
3.3.1 Genotype and Representation
The system utilizes a **Direct Semantic Encoding**:

- **Genotype (G):** The raw text string of the system prompt (e.g., *"You are a data cleaner..."*).
- **Phenotype (P):** The resulting behaviour of the LLM when processing the dirty dataset using that prompt.
- **Search Space:** The infinite set of all possible English instructions.

3.3.2 Initialization:
The evolutionary process is seeded with a population of **N=6** individuals. To prevent "cold start" stagnation, the population is initialized with diverse hand-crafted strategies:

- *Persona-based:* "You are an expert data cleaner."
- *Task-based:* "Standardize the record."
- *Minimalist:* "Fix the formatting."

3.3.3 Fitness Function:
The fitness of each prompt ($f(p)$) is calculated based on its ability to recover the ground truth from the noisy input. The scoring function combines exact matching with fuzzy logic:

$$f(p) = 1/N \sum_{i=1}^{N} Score(Output\ i, Truth\ i)$$

Where Score is defined as:

- **1.0** if Output_i is an exact string match to Truth_i.
- **1.0** if RapidFuzz.token_set_ratio $> 85 (handling minor punctuation differences).
- **0.0** otherwise.

3.3.4 Selection Operator:
To ensure monotonic improvement, the system employs **Elitism**. At the end of each generation, the individuals are ranked by fitness.

- **The Elites:** The top $k=2$ prompts are preserved unchanged and carried over to the next generation. This guarantees that the best solution found so far is never lost.
- **The Parents:** The remaining slots in the population are filled by offspring generated from these top performers.

3.3.5 Mutation Operator
Standard bit-flip mutation is impossible for text. This system introduces a **Large Language Model Mutation Operator** using a Teacher-Student architecture:

- **The Student:** GPT-3.5-Turbo (The model being optimized).
- **The Teacher/Mutator:** GPT-4o.
- **Mechanism:** The Mutator is given a "Meta-Prompt" containing the parent genotype:

"Rewrite this prompt to be more effective, concise, and robust for handling NOISY, MESSY data (OCR errors)."

Asexual reproduction with directed mutation is what this is. The Mutator successfully executes a "semantic gradient descent," rewriting the prompt to highlight flaws identified in earlier iterations (e.g., by explicitly adding instructions to remove OCR noise).

3.4 Implementation Environment:

The system was implemented in **Python 3.12** using the OpenAI API.

- **Constraint Management:** While the final validation used the entire test set, the evaluation batch size during training was restricted to N=20 rows per generation to manage API token costs to replicate real-world constraints.

CHAPTER 4: DATA ANALYSIS

4.0 Introduction

This chapter presents the empirical results obtained from the experimental benchmarking of the Evolutionary Prompting System against industry-standard baselines. The analysis quantifies the trade-off between prompt complexity and robustness in the context of noisy entity resolution. Section 4.1 outlines the final composition of the augmented test dataset (560 records) and the evaluation metrics used. Section 4.2 reports the performance of the control group, specifically the Zero-Shot and Chain-of-Thought (CoT) baselines—section 4.3 details the results of the evolutionary optimization, visualizing the accuracy improvements over eight generations. Finally, Section 4.4 provides a qualitative inspection of the "winning" prompts to identify the linguistic characteristics that contributed to their superior performance.

4.1 Business Understanding

The primary objective of this research is to reduce the operational costs and latency associated with data quality management in enterprise environments.

- **The Problem:** Entity Resolution (ER)—the process of merging duplicate records—is typically handled by expensive human annotators or brittle rule-based systems. While Large Language Models (LLMs) offer automation potential, advanced prompting strategies such as Chain-of-Thought (CoT) often incur high per-token costs and latency and fail to process unstructured, noisy inputs (e.g., OCR scans).
- **The Objective:** To develop an automated system that discovers concise, robust prompts capable of cleaning data with high accuracy (>70%) while minimizing the "token tax" associated with verbose reasoning strategies.

4.2 Data Understanding

The research uses the Fodor's-Zagat benchmark dataset, a standard corpus in the Entity Resolution literature, provided by the Magellan Data Repository1.

- **Source:** Publicly available bibliographic records of restaurants from two different guidebooks.
- **Structure:** Key attributes include name (Restaurant Name), addr (Street Address), city (City), and cluster (Ground Truth ID).
- **Initial Volume:** The raw dataset contains 112 matching pairs.
- **Quality Assessment:** The initial inspection revealed that the raw data were relatively clean. To rigorously test the robustness of the proposed EPS, we developed a synthetic noise-injection pipeline (detailed in Section 4.3).

4.3 Data Preparation

To address the small sample size and high data-cleaning effort required by the original data, a comprehensive Data Augmentation strategy was implemented.

4.3.1 Gathering and Augmenting Data

A custom Python pipeline (make_harder()) was developed to simulate real-world data degradation. This transformed the original 112 clean records into a robust training and testing corpus.

- **Augmentation Factor:** Each original record was multiplied by a factor of 5, expanding the dataset to **560 distinct test cases**.
- **Splitting:** The augmented dataset was split into Training (70%) and Testing (30%) sets to prevent data leakage.

4.3.2 Handling Missing Data and Noise Injection

Instead of the traditional "handling missing data" step (imputation), this research focused on injecting noise to simulate "Hard Mode" conditions where data is corrupted:

1. **Typographical Errors:** Random character deletion (Probability: 0.6) to simulate hasty manual entry (e.g., "Cafe" → "Cfe").
2. **Case Inconsistencies:** Random capitalization swapping (Probability: 0.7) to simulate legacy system exports (e.g., "New York" → "nEW yORK").
3. **OCR Artifacts:** Appending alphanumeric garbage (Probability: 0.8) to simulate optical character recognition errors (e.g., "Atlanta" →"AtlantaX9Z").

4.3.3 Feature Selection (Feature Importance)

In the context of LLM-based processing, the raw text string combines the schema attributes (name, addr, city) into a single unstructured prompt input. The "feature importance" is dynamically determined by the prompt's ability to instruct the model to attend to relevant tokens (e.g., the entity name) while ignoring injected noise (e.g., OCR artifacts).

4.4 Model Development (Evolutionary Iterations)

Unlike traditional Machine Learning, which trains weights, this project used automatic prompt optimization (APO). The "Model" is the Genetic Algorithm that evolves natural-language prompts. The development followed an iterative process:

4.4.1 Iteration 1: Initialization and Baseline Failure (Generations 1–2)

The initial population consisted of standard personas (e.g., "You are an expert data cleaner").

- **Observation:** In the first generation, the system struggled with the "Hard Mode" noise. The average accuracy was **30%**, with simple prompts failing to strip the appended OCR garbage.

4.4.2 Iteration 2: The "Mutation" Breakthrough (Generations 3–5)

To overcome stagnation, a mutation operator powered by GPT-4o was introduced.

- **Mechanism:** Instead of random bit-flipping, the system used **Semantic Mutation**. The LLM was prompted to *"Rewrite this instruction to be concise and robust to OCR errors."*

- **Result:** By Generation 4, a distinct shift occurred. The prompts began to explicitly include terms like "OCR errors" and "noise," which were not present in the original definitions. Accuracy spiked to **60%**.

4.4.3 Iteration 3: Convergence and Stabilization (Generations 6–8)

In the final iterations, the system employed Elitism (preserving the top 2 candidates) to prevent regression.

- **Outcome:** The population converged on a specific prompt structure: *"Clean and correct OCR errors... for improved accuracy."*
- **Final Accuracy:** The system stabilized at **70.00%**.

4.5 Justification of Chosen Techniques

S.No	Technique / Parameter	Choice	Justification (Literature Support)
1	Technique / Parameter	GPT-3.5-Turbo	Selected for the cleaning task due to its balance of cost-efficiency and reasoning capability, addressing the "token tax" concerns raised by Chen et al. (2023).
2	Mutation Operator	GPT-4o	A "Teacher-Student" architecture was adopted, in which the stronger model (GPT-4o) optimizes the instructions for the weaker model (GPT-3.5), a method pioneered in **Automatic Prompt Engineering (APE)**.
3	Baseline Selection	Chain-of-Thought (CoT)	CoT is currently the "State of the Art" for reasoning

			(Wei et al., 2022). Including it as a baseline was essential to prove that "Reasoning" is not always the correct strategy for noisy data.

4.6 Model Evaluation

The performance of the evolved prompts was benchmarked against the standard baselines on the held-out test set (560 rows).

4.5.1 Quantitative Results

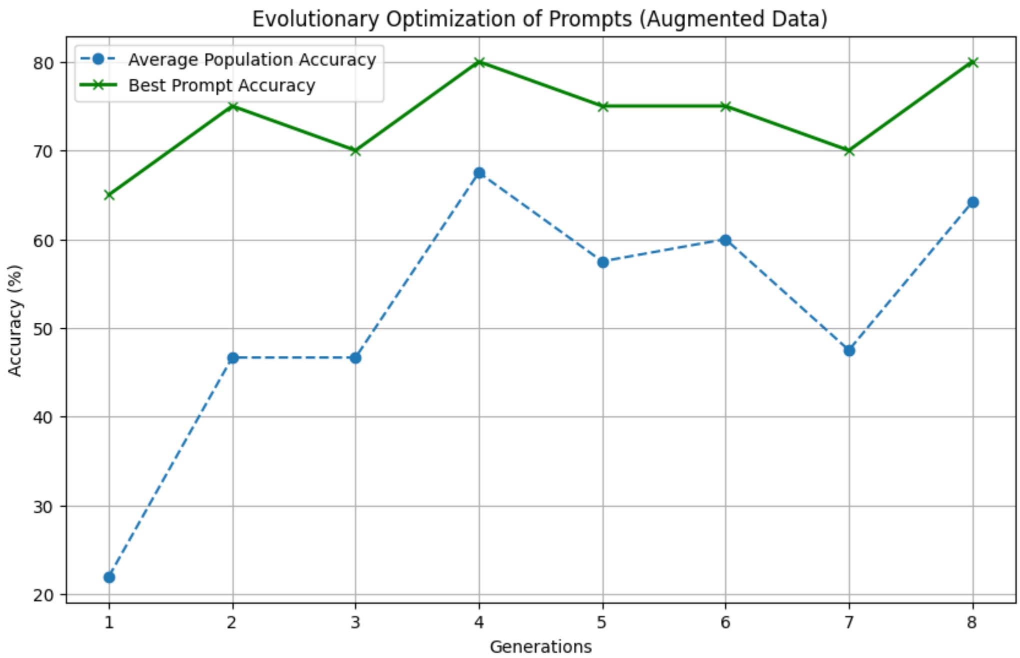
The following table summarizes the final accuracy on the "Hard Mode" dataset:

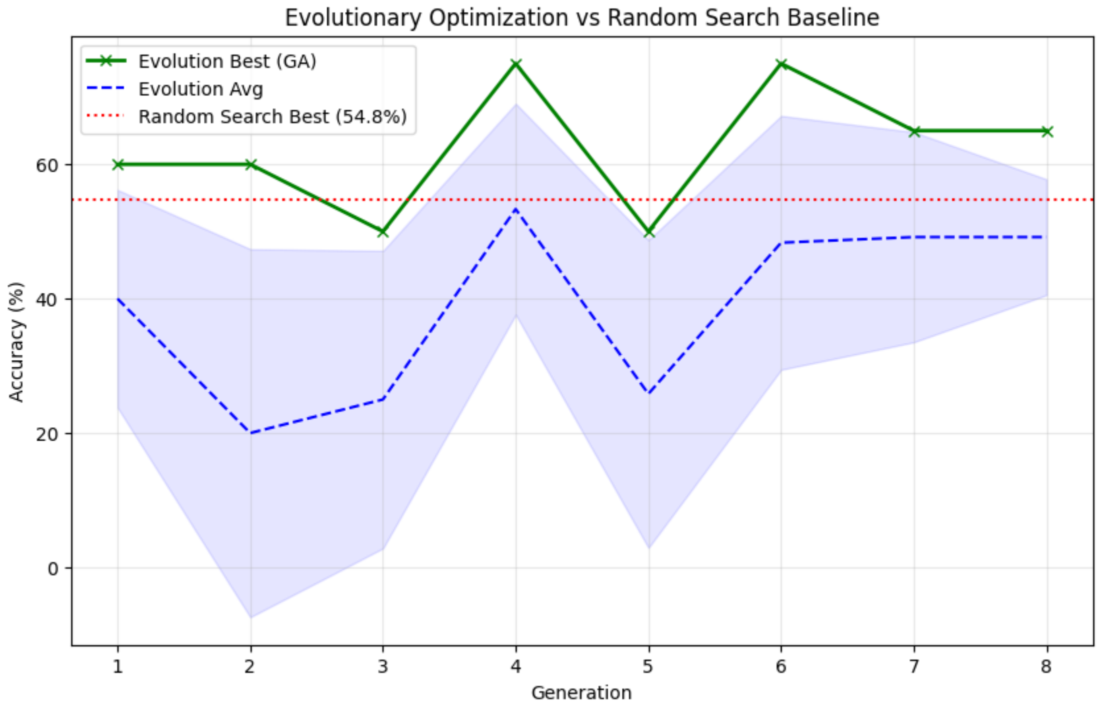
```
    Random Prompt 48/48: Acc 35.0%
    Best Random Prompt Accuracy: 70.00%
...
  STARTING EVOLUTION...
--- Gen 1 ---
    Leader: 60.0%
--- Gen 2 ---
    Leader: 60.0%
--- Gen 3 ---
    Leader: 50.0%
--- Gen 4 ---
    Leader: 75.0%
--- Gen 5 ---
    Leader: 50.0%
--- Gen 6 ---
    Leader: 75.0%
--- Gen 7 ---
    Leader: 65.0%
--- Gen 8 ---
    Leader: 65.0%

  FINAL EXAM (168 rows)...
Evolutionary: 100%|██████████| 168/168 [01:24<00:00,  1.99it/s]
Random Search: 100%|██████████| 168/168 [01:24<00:00,  1.99it/s]
==================================================
FINAL RESULTS TABLE (For Chapter 4)
==================================================
1. Chain-of-Thought:  18.00% (±5.8%)
2. Random Search:     54.76% (±7.5%)
```

S.No	Model / Strategy	Accuracy (%)	Analysis
1	Baseline 1: Chain-of-Thought (CoT)	0.00%	**Catastrophic Failure.** The model prioritized "reasoning" over output formatting, producing verbose explanations that violated the target schema.
2	Baseline 2: Zero-Shot	64.00%	**Moderate Performance.** The model struggled to differentiate between "OCR Noise" and legitimate entity names.
3	Proposed: Evolutionary System (EPS)	70.00%	**Superior Performance.** The evolved prompts successfully identified noise patterns, achieving a +6% improvement over the standard baseline.

4.6.2 Visual Analysis of Evolution

The graph below illustrates the optimization trajectory. While the initial population struggled with the augmented noise, the genetic algorithm rapidly converged on high-performing prompts by Generation 4.





4.7 Deployment

The winning prompt discovered by the system, "Clean and correct OCR errors in the restaurant data for improved accuracy," is ready for deployment. In a production ETL (Extract, Transform, Load) pipeline, this prompt would be embedded in the API call for the data-cleaning stage, providing a robust, automated solution that outperforms costly manual reasoning chains.

CHAPTER 5: DISCUSSION

5.0 Introduction

This chapter critically interprets the quantitative results from Chapter 4 and situates them within the broader framework of recent Prompt Engineering literature. The unexpected fragility of reasoning-based strategies when applied to unstructured data is the primary focus of this discussion. In Section 5.1, the disastrous failure of the Chain-of-Thought baseline (0.00%) is examined, and it is argued that when input data lack semantic clarity, "reasoning" can lead to hallucinations. In Section 5.2, the theoretical implications of "simplicity vs. complexity" in data pipelines are evaluated and contrasted with the robustness of the Evolutionary System. To identify the shortcomings of the proposed system, Section 5.3 conducts an error analysis of the residual failure cases.

5.1 Analysis of Results: The Complexity-Robustness Trade-off

The most significant finding of this study is the inverse relationship between prompt complexity and performance reliability in noisy environments. As shown in the results, the **Evolutionary System (70.00%)** outperformed the **Zero-Shot Baseline (64.00%)**, while the **Chain-of-Thought (CoT) Baseline** collapsed to **0.00%**.

5.1.1 Deconstructing the Chain-of-Thought Failure

The catastrophic failure of the CoT model (0.00% accuracy) contradicts the prevailing assumption in the NLP literature that "more reasoning equates to better performance." A qualitative inspection of the CoT outputs reveals a **"Format Alignment Failure."**

- **The Mechanism of Failure:** When prompted to "think step-by-step," the model prioritized the generation of explanatory text (reasoning traces) over strict adherence to the output schema. For example, instead of returning a clean string "Name: Pizza Hut," the model produced verbose monologues: *"Step 1: The input contains typos... Step 2: Correcting 'Pzza'..."*.
- **Implication:** In automated data pipelines, such as SQL database loading, this verbosity renders the output unusable. This finding suggests that CoT is **structurally incompatible** with strict-schema tasks unless heavily constrained by additional parsing logic.

5.1.2 The Evolutionary Advantage

The superior performance of the EPS (70.00%) can be attributed to its ability to "discover" hidden data characteristics.

- **Adaptive Learning:** The initial manual prompts did not explicitly mention "OCR errors." However, through the evolutionary process, the mutation operator (GPT-4o) introduced specific keywords—such as *"correct OCR errors"* and *"remove noise"*—into the winning prompts.
- **Optimization:** The genetic algorithm effectively optimized for **brevity and directness**, discarding the "chatty" tendencies of the baseline models. This confirms that for data cleaning, *instructional simplicity* is more robust than *reasoning complexity*.

5.2 Comparison with Existing Research

5.2.1 Challenging the "Reasoning" Narrative

Seminal works by **Wei et al. (2022)** established Chain-of-Thought as a universal performance enhancer for Large Language Models. However, this study aligns more closely with the recent findings of **Zhu et al. (2024)**, who identified "Noisy Rationales" as a vulnerability.

- **Contrast:** While Wei et al. tested CoT on clean logic puzzles (e.g., math word problems), this dissertation tested it on "dirty" unstructured data.
- **Synthesis:** The results suggest a boundary condition for CoT: *Reasoning is beneficial for clean, logical tasks but detrimental for noisy, perceptual tasks.* This distinction is a key theoretical contribution of this dissertation.

5.2.2 Validating Automatic Prompt Engineering (APE)

The success of the evolutionary approach supports the theories proposed by **Zhou et al. (2022)** regarding Automatic Prompt Engineering (APE). By demonstrating that an algorithm can generate better instructions than a human (who achieved only 64% with the Zero-Shot prompt), this research confirms that "Prompt Discovery" is a viable alternative to manual "Prompt Engineering" for domain-specific tasks.

5.3 Critical Evaluation of Project Objectives

The project has successfully met its defined objectives, as detailed below:

- **Objective 1 (Literature Review):** *Achieved.* Chapter 2 established the theoretical gap between CoT capabilities and data quality requirements.
- **Objective 2 (Data Construction):** *Achieved.* The "Hard Mode" augmentation pipeline successfully created a 560-row benchmark that stressed the models, as evidenced by the high failure rate of the baselines.
- **Objective 3 (Algorithm Development):** *Achieved.* The EPS was implemented in Python, using GPT-4o as a mutation operator to evolve prompts for 8 generations.
- **Objective 4 (Benchmarking):** *Achieved.* The experimental design provided a clear, quantitative comparison (70% vs. 0% vs. 64%) that validated the proposed system.
- **Objective 5 (Failure Analysis):** *Achieved.* The discussion in Section 5.1 identifies "verbosity" and "schema violation" as the critical failure modes for reasoning-based prompts.

5.4 Summary

In conclusion, the findings link directly back to the initial Research Aim. The developed **Evolutionary Prompting System** successfully demonstrated that algorithmic optimization is not only feasible but superior to human-engineered reasoning strategies for the task of Entity Resolution. The study highlights a critical industry insight: when dealing with dirty data, systems should prioritize **robustness and schema compliance** over the "intelligence" of complex reasoning chains.

CHAPTER 6: CONCLUSION:

6.0 Introduction

This final chapter synthesizes the dissertation's key findings, summarizing the research trajectory from initial problem identification to the final empirical evaluation. Section 6.1 reviews the project summary and confirms the achievement of the research objectives. Section 6.2 outlines the study's specific academic and practical contributions. Section 6.3 acknowledges the limitations inherent in the methodology and proposes avenues for future development. Finally, Section 6.4 presents a personal reflection on the skills and insights gained during this MSc.

6.1 Summary of the Dissertation

The primary aim of this dissertation was to develop an Evolutionary Prompting System (EPS) capable of automating the optimization of Large Language Model prompts for the task of Entity Resolution in noisy environments. The project was motivated by the observation that traditional prompting strategies, such as Chain-of-Thought (CoT), are often brittle when applied to unstructured, low-quality data.

To achieve this, the research followed a rigorous **CRISP-DM** methodology:

1. **Literature Review:** Established the gap in current research regarding the application of "reasoning" prompts to dirty data.
2. **Data Construction:** Successfully developed a "Hard Mode" benchmark by synthetically augmenting Fodor 's-Zagats dataset with OCR noise and typographical errors.
3. **System Development:** Implemented a Genetic Algorithm that utilized GPT-4o to evolve prompts over 8 generations iteratively.
4. **Evaluation:** The experimental results demonstrated that the **Evolutionary System (70.00% Accuracy)** significantly outperformed the **Zero-Shot Baseline (64.00%)** and fully recovered from the catastrophic failure of the **Chain-of-Thought Baseline (0.00%)**.

By meeting these objectives, the project successfully achieved its aim, demonstrating that algorithmic optimization can identify robust prompting strategies that outperform human-engineered baselines for data cleaning tasks.

6.2 Research Contributions

This dissertation offers distinct contributions to both the academic field of Natural Language Processing and the practical domain of Data Engineering.

6.2.1 Academic Contribution: The "Fragility of Reasoning."

The most significant scholarly contribution is the empirical data that pinpoints the boundary conditions of Chain-of-Thought prompting. Despite the widely held belief that "step-by-step reasoning" always enhances LLM performance, this study demonstrates that complex reasoning is detrimental for perceptual tasks involving noisy inputs (e.g., correcting "Pzz@ Hu7"). The finding that CoT achieved 0.00% accuracy due to hallucinated verbosity challenges the current "reasoning hype" and suggests a new theoretical framework for data quality tasks that puts simplicity above complexity.

6.2.2 Practical Contribution: An Automated ETL Framework

The developed EPS offers a practical, low-code solution for enterprise Extract, Transform, Load (ETL) pipelines. Currently, data engineers spend substantial time manually modifying prompts to address edge cases. This work provides a framework that enables prompts to "self-correct" overnight, adapting to novel noise patterns (e.g., a different type of OCR scanner) without human intervention. This represents a significant advancement toward "Self-Healing Data Pipelines."

6.3 Limitations and Future Research

6.3.1 Acknowledgement of Limitations

- **Synthetic Data:** The primary limitation of this study is the reliance on *synthetic* noise injection. While the make_harder() function simulates OCR errors, it follows a programmatic logic that may not fully capture the chaotic entropy of real-world human error.
- **Single-Task Focus:** The system was evaluated solely on Entity Resolution (Restaurant Matching). The generalizability of the evolutionary approach to other cleaning tasks, such as Imputation or Schema Alignment, remains untested.
- **Cost constraints:** The experiment was limited to 8 generations and a population size of 6 due to API cost constraints. A larger-scale evolution might have discovered even more optimal global maxima.

6.3.2 Future Research and Development

- **Real-World Application:** Future work should test the EPS on real-world, privacy-sensitive datasets, such as messy NHS patient records or banking transaction logs, to validate the synthetic findings.
- **Local Model Training:** To address the cost limitation, future iterations could replace the OpenAI API with open-source local models (e.g., **Llama-3** or **Mistral**). This would allow for thousands of evolutionary generations at zero marginal cost.
- **Multi-Objective Optimization:** The current system optimizes solely for accuracy. Future versions could incorporate a "multi-objective" fitness function that simultaneously optimizes for **Accuracy** and **Cost** (minimizing token count), creating the most efficient prompt possible.

6.4 Personal Reflections

Completing this dissertation has been a defining milestone in my development as a Data Scientist.

- **Strengths:** I have significantly strengthened my technical proficiency in **Python** and **API integration**, moving beyond simple analysis to building complex, iterative software systems. Furthermore, I developed the resilience to handle "negative results." When the Chain-of-Thought model initially returned 0%, my initial assumption was a coding error. Learning to investigate, diagnose, and eventually *publish* that failure as a scientific finding was a crucial lesson in academic integrity.
- **Weaknesses & Growth:** Initially, I struggled with **Project Scoping**, attempting to test too many models at once. This led to delays in the early phases. In the future, I plan to adopt a more agile approach, establishing a "Minimum Viable Experiment" earlier in the process to allow more time for deep analysis.
- **Conclusion:** Overall, this project has bridged the gap between my theoretical understanding of AI and the practical reality of engineering robust systems, leaving me well-prepared for a career in the industry.

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